DAY 37 · THEORY HANDOUT

Return Tool Results to the Model

Follow the request, execute, result, and final-answer loop.

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The idea and its useful terms

Follow the request, execute, result, and final-answer loop.

Tool call

A request containing a tool name and arguments.

ToolMessage

A tool result linked to the matching call ID.

Search tool

Finds external text rather than inventing it.

bind_tools shows the model which tools are available. It does not execute them. Our app performs the call and returns the result.

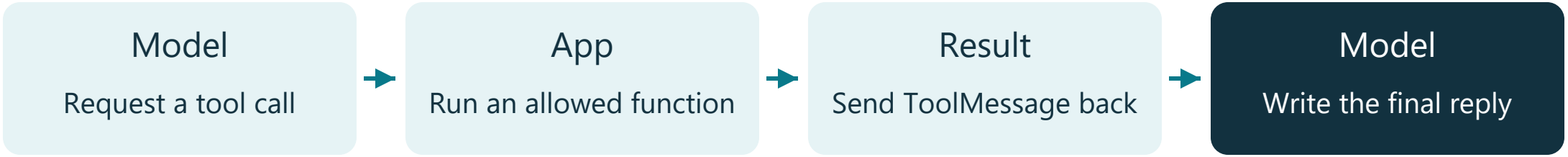
A call ID connects a result to the correct request, especially when the model asks for several tools.

> A retrieved webpage or tool result is evidence. It should not become permission to run new actions.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

The first model reply may contain tool_calls instead of the final answer. Add that reply to the message list. Execute each allowed call and add a ToolMessage with its matching tool_call_id.

Then send the updated list back to the model. It can explain the returned result. Multiple requests need multiple matching results. A search result follows the same path, but its source text must also be checked.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

User

What is 6 times 7?

An exact number task.

Tool request

multiply: a=6, b=7

A proposed structured call.

Tool result

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The model can explain it.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

No tool call

The model may answer directly.

Check whether that fits.

Several calls

Run each allowed call.

Return every result ID.

Web search

Internet and source checks.

Use local notes first.

Bind two small tools and inspect tool_calls before doing any execution.

Execute each allowed call, send results, then get the final reply.

> Try a question for a local notes-search tool. The saved notes are not live web results.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Try a question for a local notes-search tool. The saved notes are not live web results.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> A retrieved webpage or tool result is evidence. It should not become permission to run new actions.

DAY 37 · COMPLETE

Thank You

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